

Lesson 4: Python, Data Preparation and Classical ML Foundations

Lesson 4: Python, Data Preparation and Classical ML Foundations NumPy, pandas, preprocessing, classification, and evaluation

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Learning Outcomes

- Use NumPy arrays and understand shape.
- Create and inspect a pandas DataFrame.
- Split data into training and testing sets.
- Handle missing values, encode categories and scale numeric features.
- Train a KNN classifier and interpret accuracy/confusion matrix.

Lesson 4 at a Glance

- Part A — NumPy: arrays, reshape, arithmetic, random matrices.
- Part B — pandas: tabular data and filtering.
- Part C — Logistic Regression: a train/test workflow.
- Part D — Preprocessing: imputation, one-hot encoding, scaling.
- Part E — KNN: similarity-based classification and evaluation.

Part A — NumPy Foundations

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5])
print("Array:", arr)
matrix = arr.reshape(5, 1)
print("Reshaped Matrix:\n", matrix)
arr_squared = arr ** 2
print("Squared Array:", arr_squared)
random_matrix = np.random.rand(3, 3)
print("Random Matrix:\n", random_matrix)
```

- `np.array` creates a numerical array.
- `reshape(5,1)` changes the shape to five rows and one column.
- `arr ** 2` squares each element element-wise.
- `np.random.rand(3,3)` creates random values in a 3×3 matrix.

Interpreting the NumPy Output

```
Array: [1 2 3 4 5]
```

- The reshaped matrix is a column vector.
- The squared array is [1, 4, 9, 16, 25].
- The random matrix changes between runs because it is generated randomly.

Part B — pandas DataFrames

```
import pandas as pd
data = {'Name': ['Alice', 'Bob', 'Charlie'],
        'Age': [25, 30, 35],
        'Salary': [50000, 60000, 70000]}
df = pd.DataFrame(data)
print(df.head())
print(df.describe())
df.fillna(0, inplace=True)
filtered_df = df[df["Age"] > 30]
print(filtered_df)
```

- A DataFrame organizes rows and columns.
- `head()` previews the data.
- `describe()` summarizes numerical columns.
- Boolean filtering keeps rows satisfying a condition.

Interpreting the pandas Output

- The dataset contains three people.
- The mean age is 30 and mean salary is 60,000.
- Filtering Age > 30 keeps Charlie.
- This is the tabular-data foundation used by many ML workflows.

Part C — Logistic Regression Workflow

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

X = [[1], [2], [3], [4], [5]] * 4
y = [1,2,3,4,5,1,3,3,4,5,1,2,3,4,5,1,1,3,4,5]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

- The notebook uses a supervised multiclass classification example.
- `train_test_split` separates training and testing examples.
- `fit` learns from the training data.
- `predict` generates class labels for the test data.

Accuracy: 0.75

- 75% of the held-out test predictions were correct in the recorded run.
- Accuracy depends on the selected split and data.
- The example demonstrates the standard train → predict → evaluate workflow.

Part D — Preparing Data for Learning

```
data = {'Age': [25, 30, 35, 40, None],  
        'Salary': [50000, 60000, 70000, None, 90000],  
        'Gender': ['Male', 'Female', 'Male', 'Female', 'Male']}  
df = pd.DataFrame(data)  
df["Age"] = df["Age"].fillna(df["Age"].mean())  
df["Salary"] = df["Salary"].fillna(df["Salary"].mean())  
encoder = OneHotEncoder(sparse_output=False)  
gender_encoded = encoder.fit_transform(df[['Gender']])  
scaler = StandardScaler()  
df[['Age', 'Salary']] = scaler.fit_transform(df[['Age', 'Salary']])
```

- Missing values are filled using the mean of the numeric column.
- One-hot encoding turns a category such as Gender into numerical columns.
- StandardScaler transforms numerical features to a standardized scale.
- Quantum ML pipelines also need carefully prepared numerical inputs.

Interpreting the Preprocessed Output

Age Salary Male Female

- Age and Salary are centered/scaled numeric features.
- Male/Female are represented as indicator columns.
- The output is suitable for a numerical ML algorithm.

Part E — KNN Classification

```
X = df[['Age', 'Salary']]
y = df['Purchased']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print(f"Model Accuracy: {accuracy * 100:.2f}%")
print("Confusion Matrix:\n", conf_matrix)
```

- KNN compares a new point with nearby training points.
- `n_neighbors=3` uses the three nearest neighbors.
- Standardization prevents Age and Salary from being compared on very different scales.
- The final evaluation uses accuracy and a confusion matrix.

Model Accuracy: 100.00%

$[[1,0],[0,1]]$

- The recorded test examples were all classified correctly.
- The diagonal confusion-matrix entries are correct predictions.
- A perfect score on a tiny dataset should not be interpreted as proof of general performance.

Why This Lesson Matters for Quantum ML

- Quantum ML still needs clean, numeric, appropriately scaled input data.
- The later feature maps encode classical numbers into quantum states.
- Model evaluation remains classical: accuracy, test predictions and error analysis still matter.
- Good preprocessing is a prerequisite—not an optional extra.

Beginner Activity

- Change the threshold in the pandas filter.
- Try a different train/test split.
- Change KNN from 3 to 5 neighbors.
- Discuss how scaling affects distance-based KNN.

Lesson Summary

- NumPy handles numerical structure.
- pandas handles tabular data.
- Scikit-learn supplies modeling and evaluation.
- Preprocessing converts heterogeneous data into usable numerical features.

Data preparation is the bridge between raw information and a trainable quantum or classical model.

Quick Check

- ① Why reshape a vector?
- ② What does one-hot encoding accomplish?
- ③ Why scale Age and Salary?
- ④ What does a confusion matrix tell us?
- ⑤ Why should test accuracy be interpreted carefully?